Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A wall is 50 m away. An echo returns after 0.5 s. A student divides one-way distance by this time. Find the correct speed and explain the student's error.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; measure angles from the normal. The image is virtual.

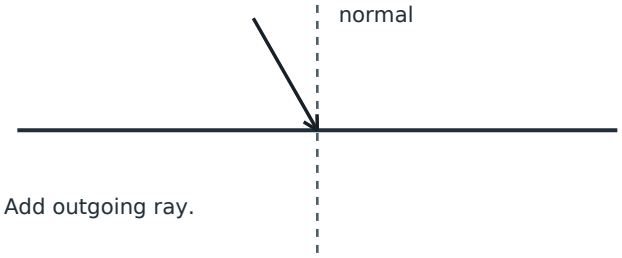
Worked reasoning: An object 2 m in front of an ideal plane mirror has a virtual image 2 m behind it. The object-image separation is 4 m; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

Marzano check - circle: ① ② ③

03 Look and explain

Draw the normal at the mirror. Complete a reflected ray for an incident ray 30° from normal.



Marzano check - circle: ① ② ③

04 Build the solution

An object is 2 m in front of a plane mirror. Where is its pictured image relative to the mirror? How far apart are object and image?

Marzano check - circle: ① ② ③

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05 Use the idea

Why is the mirror image called virtual rather than a real object behind the glass?

Marzano check - circle: 1 2 3

06 Choose a method

A reflected signal returns after 4 s. Name the extra quantity needed to calculate distance and explain why this is a round trip.

Marzano check - circle: 1 2 3

07 Exit check

Incidence angle 45° from normal. Find reflection angle.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

Move the object 0.5 m closer to the mirror. Find the new image distance and object-image separation.

Marzano check - circle: 1 2 3